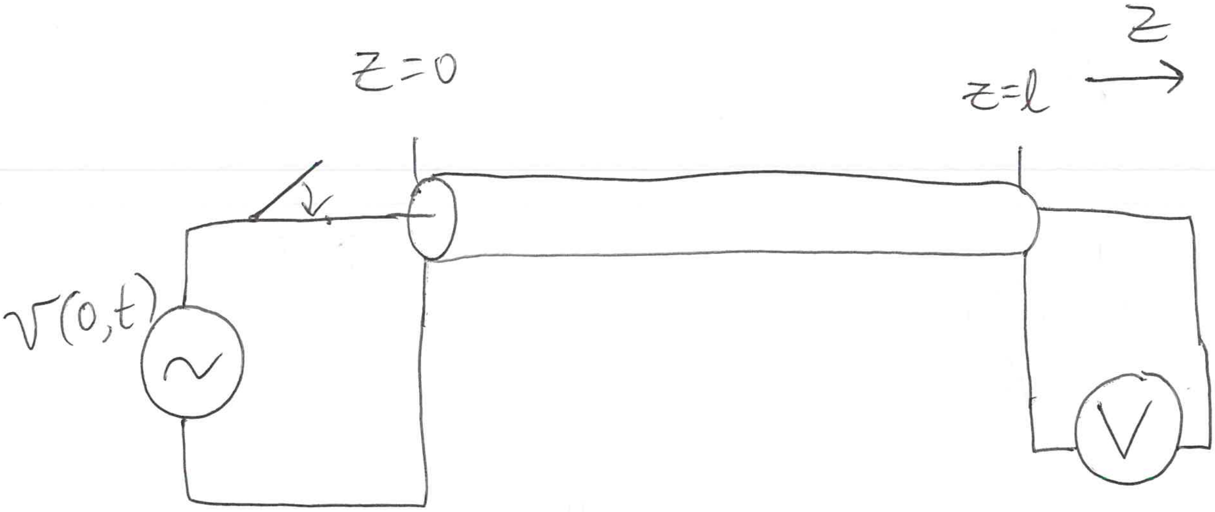
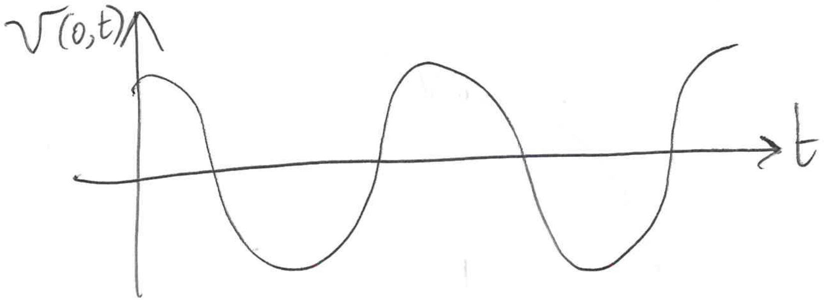
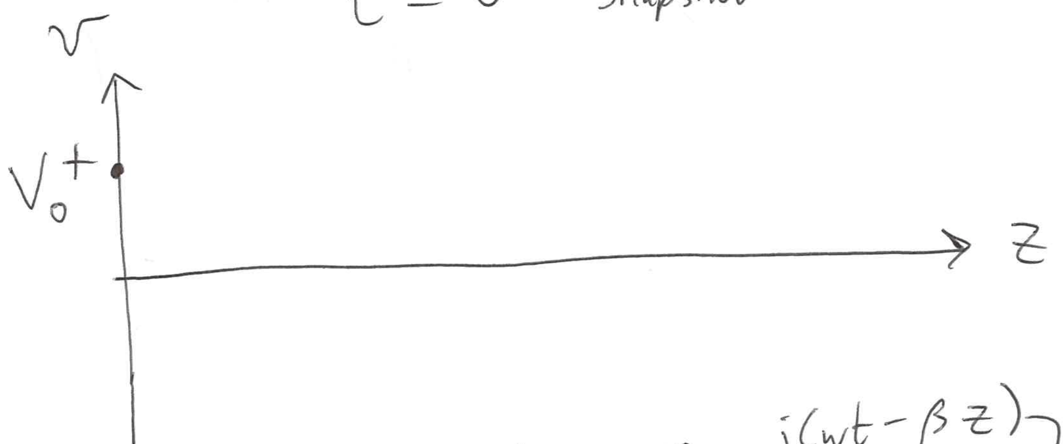




$t = 0$ Snapshot

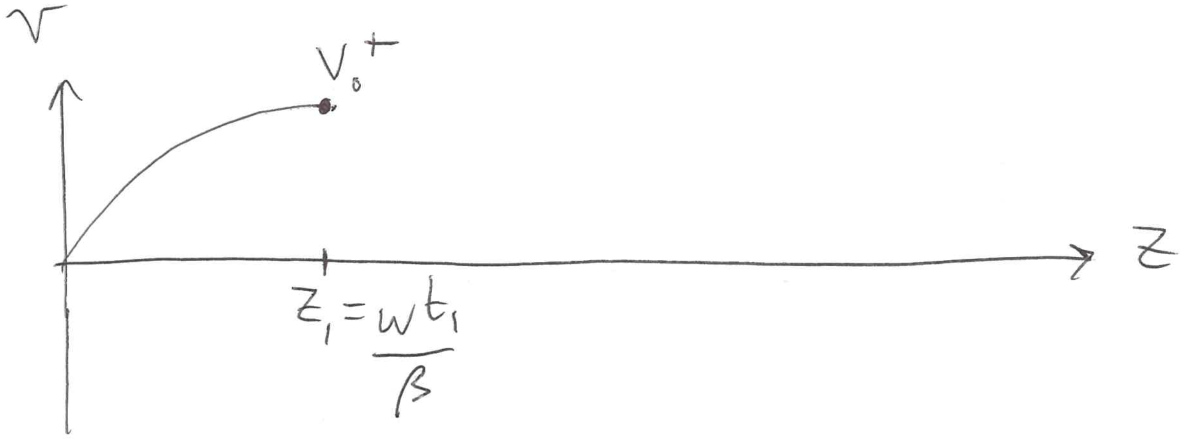


$$v(z,t) = V_0^+ \, \mathrm{Re}\left\{ e^{j(\omega t - \beta z)} \right\}$$

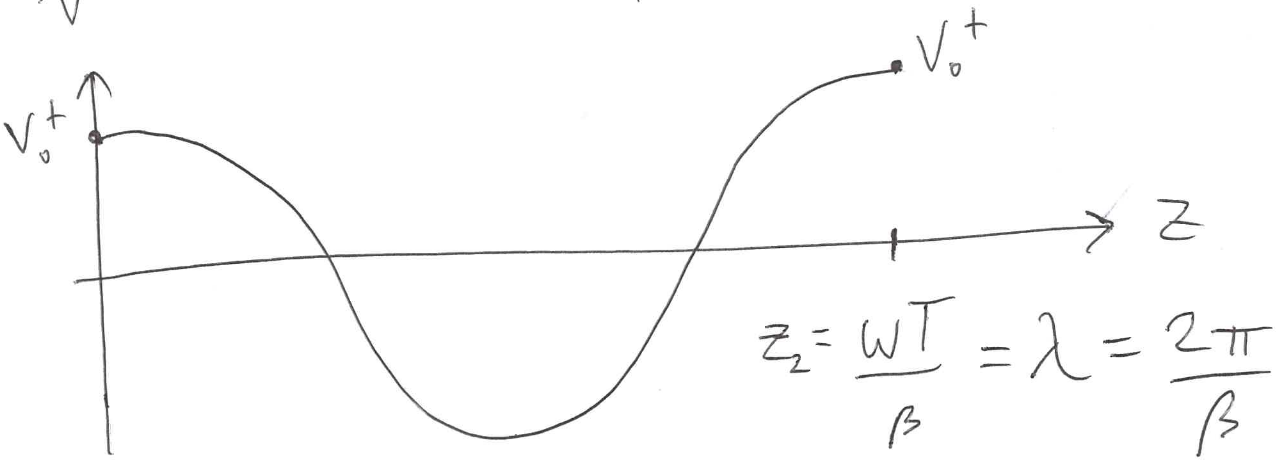
$$v(z,t) = V_0^+ \quad \text{when} \quad \omega t = \beta z \Rightarrow z = \frac{\omega t}{\beta}$$

$$v\left(\frac{\omega t}{\beta}, t\right) = V_0^+$$

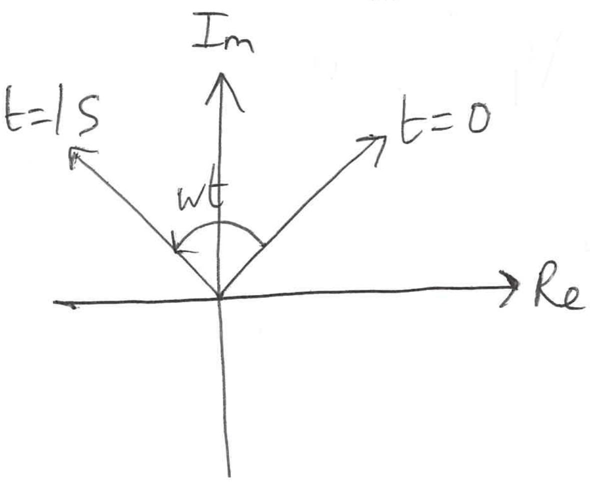
$t = t_1$ snapshot

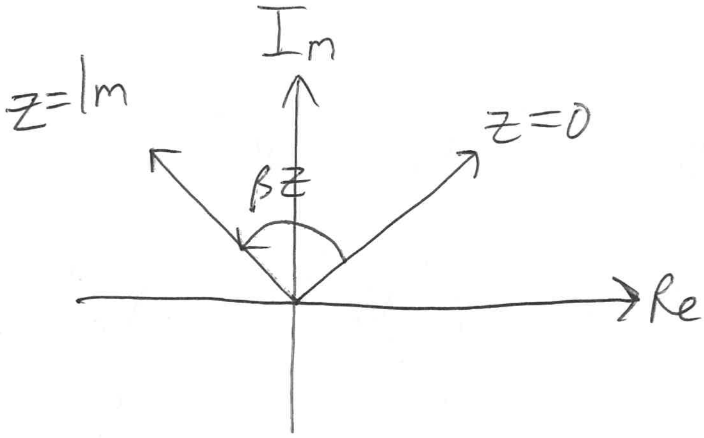


$t_2 = T = 1/f = \frac{2\pi}{w}$ snapshot

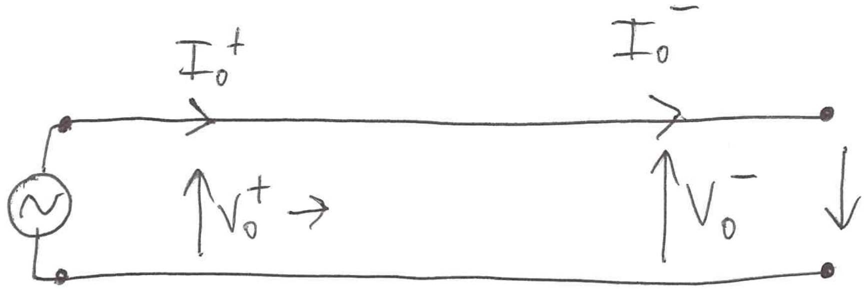


Time	Space
Angular freq. ω [rad/s]	Propagation constant $\beta = \mathrm{Im}\{\gamma\}$ [rad/m]
Period $T = \frac{2\pi}{\omega}$ [s]	Wavelength $\lambda = \frac{2\pi}{\beta}$ [m]





Open line: $Z_L = \infty \Rightarrow \Gamma = 1$



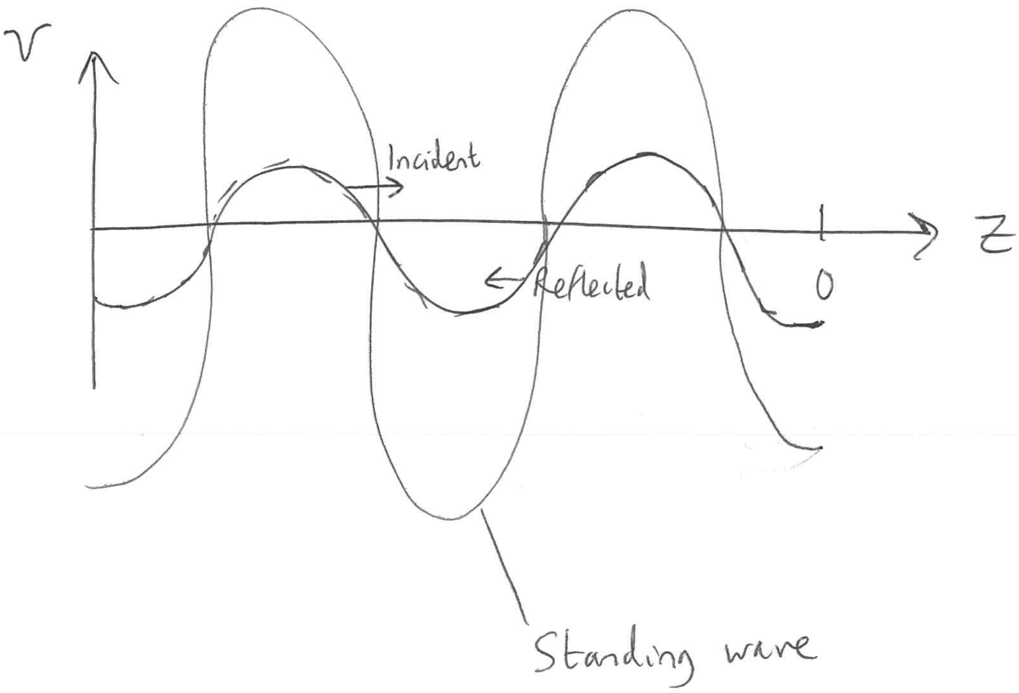
$$I_L = 0 = \frac{V_0^+ - V_0^-}{Z_0}$$

$$V_0^+ = V_0^-$$

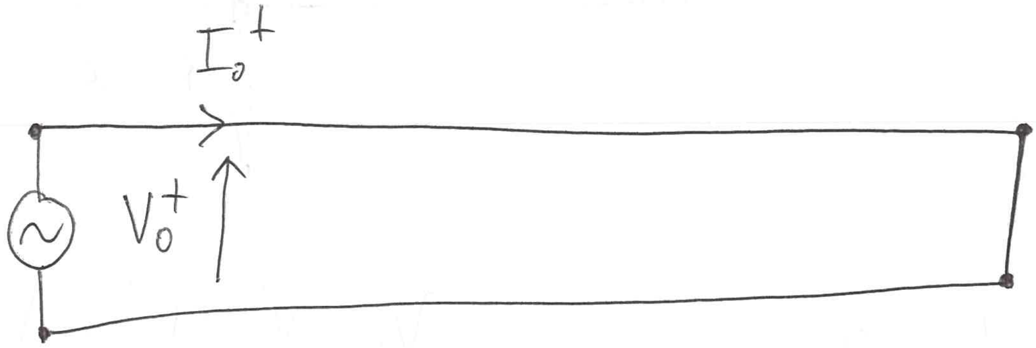
$$V_L = 2V_0^+$$

$$V_0^- = \Gamma V_0^+ = V_0^+$$

$$I_0^- = -\Gamma \frac{V_0^+}{Z_0} = -\frac{V_0^+}{Z_0} = -I_0^+$$



Short circuit: $Z_L = 0$: $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = -1$

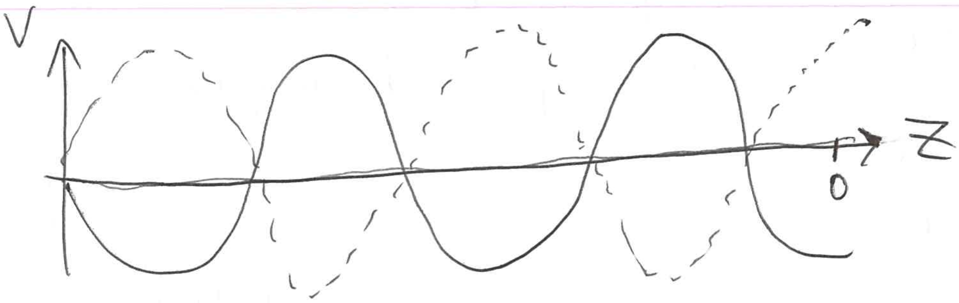


$V_L = 0 = V_0^+ + V_0^-$

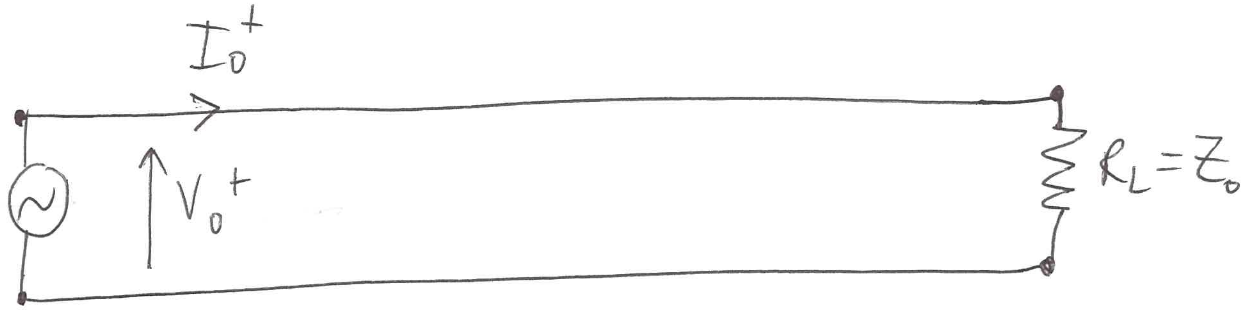
$V_0^- = \Gamma V_0^+ = -V_0^+$

$I_0^- = -\frac{\Gamma V_0^+}{Z_0} = \frac{V_0^+}{Z_0} = I_0^+$

$I_L = I_0^+ + I_0^- = 2I_0^+$



Matched line: $Z_L = Z_0 \quad \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = 0$



$V_0^- = \Gamma V_0^+ = 0$

$I_0^- = -\Gamma \frac{V_0^+}{Z_0} = 0$

$V_L = V_0^+ + V_0^- = V_0^+$

$I_L = I_0^+ + I_0^- = I_0^+$

$R_L = \frac{V_L}{I_L} = \frac{V_0^+}{I_0^+} = Z_0$

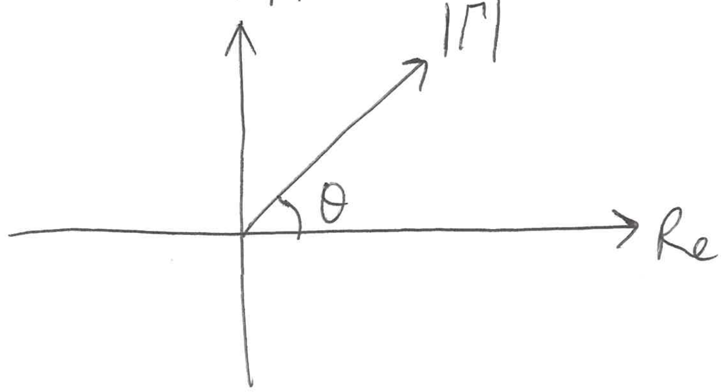
$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$R = G = 0 \Rightarrow Z_0 = \text{real}$

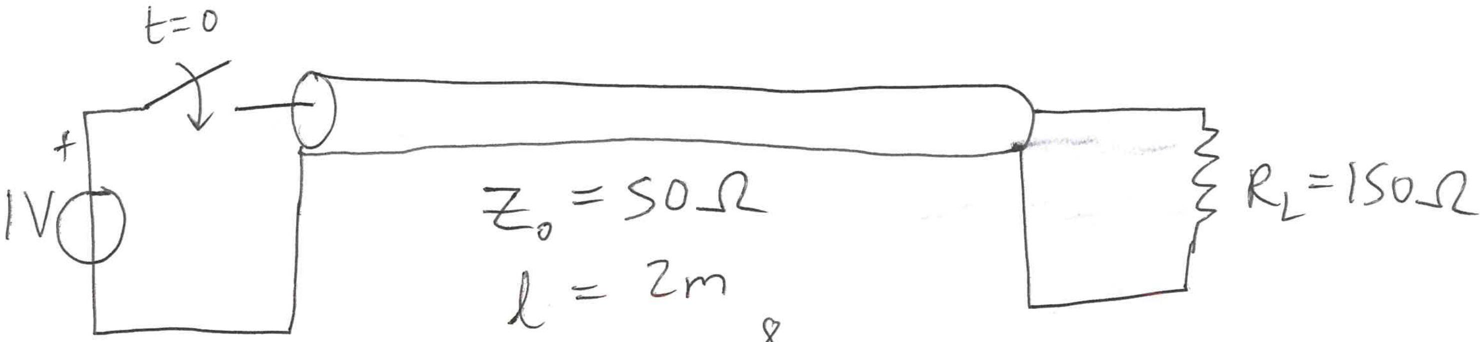
Z_L can be complex

$$\Gamma = |\Gamma| e^{j\theta}$$

$0 \leq |\Gamma| \leq 1$



$R_G = 0\,\Omega$



$Z_0 = 50\,\Omega$

$l = 2\,m$

$v_p = 2 \times 10^8\,m/s$

$$t_p = \frac{l}{v_p} = 10\,ns$$

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} = \frac{100}{200} = \frac{1}{2}$$

Whenever: $Z_L > Z_0$: Γ_L is +ive V reflected in phase

$Z_L < Z_0$: Γ_L is -ive V reflected out of phase

$$\Gamma_G = \frac{R_G - Z_0}{R_G + Z_0} = -1$$